

Covariance-horizontal synthesis, stationary low-chaos cancellation,

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-29 · this version issued 2026-07-29 · v1.0

1. Purpose and scope

R-119 isolated four questions after the legal two-block/two-visit chart had been reconstructed: whether the realized covariance quotient controls the physical endpoint norms, whether the R-118 quadratic multiplier theorem survives spatially varying matrices, whether the production low chaoses actually cancel, and whether the terminal Hessian reduces to fixed selfadjoint quadratic forms. This note attacks those questions in five coordinated rounds.

1. diagonalize the complete six-current A1 coefficient in the fixed six-real Pauli frame;
2. identify the covariance quotient with an H^2 endpoint bound and an exact L^6 bridge;
3. extend the R-118 multiplier estimate through two spatial derivatives;
4. evaluate zero and first chaos for both the one-pair diagnostic and the complete uncontrolled stationary six-row A1 raw-current packet;
5. differentiate the linear and rational terminal currents, flatten the zeroth-order rational Hessian, and audit the surviving Cartan owner.

R-120 conclusion. The covariance-horizontal synthesis problem from R-119 is solved. For every temporally faithful R-104 chart, the realized quotient norm controls the H^2 endpoint with the A8 coercivity constant and controls the endpoint L^6 difference with the sharp two-term constant 32. The source energy and the terminal sextic can therefore be spent once in the R-118 estimate. The R-118 estimate also extends to spatially varying selfadjoint matrices whose summed bilinear derivatives through order two are bounded. At $\kappa = 1/10$ its optimized remainder has the exact coefficient powers $K_0^{3/2} K_{\leq 2}^{11/6}$.

At every fixed cutoff and positive floor, the complete uncontrolled stationary six-row A1 raw-current packet satisfies both R-119 zero/first-chaos identities. This includes all three linear and all three rational current rows, but not a trace/heat/low/R-063 forest or adapted future feedback. The linear rows have an exact square-Laplacian-trace Hessian and no Cartan curvature. The rational raw-covariance Hessian has an exact selfadjoint formula and a fixed 21-matrix flattening, but the rational current retains a genuine first-order Cartan block. The required companion curl +40/729 has not been observed. The adapted coefficients D_0, D_1 , complete trace/R-063 forest reconstruction, Cartan cancellation or absorption, one-use global production inequality, OVERLAP_{src}, Nelson, removals, interacting measure, and Sector-A closure remain open. Tier T4 is unchanged.

All positive statements below are finite-cutoff, fixed-positive-floor statements in the A1 six-real convention unless a theorem explicitly states an abstract Hilbert-space form.

2. Round one: exact six-current production frame

Let $S_A = S_A^*$, $A = 1, 2, 3$, be the fixed six-real embedded Pauli generators, and write

$$m_A(u) = u^T S_A u, \quad \rho(u) = u^T u, \quad d = \rho + e, \quad q_A = \frac{m_A}{d},$$

where $e > 0$ is the rational floor. The A1 coefficient block is built from the two one-forms

$$J_A = dm_A, \quad K_A = dm_A - q_A d\rho.$$

Its scalar weights are

$$a = \frac{9}{500P}, \quad b = \frac{3}{400P}, \quad c = \frac{3}{320P}, \quad P = 4 + 10^{-12}.$$

Theorem 2.1 (exact positive diagonalization)

Set

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha = \frac{5}{9},$$

and

$$L_A = (1 - \alpha)J_A + \alpha K_A = dm_A - \alpha q_A d\rho.$$

Then, identically,

$$aJ_A^2 + 2bJ_A K_A + cK_A^2 = c_0 J_A^2 + c_1 L_A^2. \quad (2.1)$$

Indeed,

$$c_0 + c_1(1 - \alpha)^2 = a, \quad c_1\alpha(1 - \alpha) = b, \quad c_1\alpha^2 = c. \quad (2.2)$$

Consequently the complete covariance coefficient is

$$B(u) = \sum_{A=1}^3 \left(c_0 p_A p_A^T + c_1 \ell_A \ell_A^T \right), \quad (2.3)$$

where

$$p_A = 2S_A u, \quad \ell_A = 2(S_A - \alpha q_A I)u. \quad (2.4)$$

This is an exact six-current factorization; it is not a scalar-row proxy.

The fixed generators obey $S_A^2 = P_{\text{dbl}}$ on the active doublet and are selfadjoint. Hence

$$\sum_{A=1}^3 |S_A| = 3P_{\text{dbl}} \preceq 3I, \quad (2.5)$$

and the complete linear-row envelope is

$$4(c_0 + c_1) = \frac{339}{2000P}. \quad (2.6)$$

These constants are derived from the A1 manifest rather than inserted from a downstream fit.

3. Round two: covariance-horizontal endpoint synthesis

Let $\mathcal{S} = \bigoplus_k \mathcal{S}_k$ be the source Cameron–Martin space of a fixed R-104/R-093 chart and let

$$L_\pi h = \sum_k S_k h_k, \quad C_J = L_\pi L_\pi^* = \sum_k S_k S_k^*. \quad (3.1)$$

For $z \in \text{Ran } L_\pi$, define the realized quotient norm

$$\|z\|_{q, L_\pi}^2 = \|C_J^{\dagger/2} z\|^2. \quad (3.2)$$

Theorem 3.1 (lossless quotient and physical H^2 control)

For every source realization h and $z = L_\pi h$,

$$\|z\|_{q,L_\pi}^2 = \|P_{(\text{Ker } L_\pi)^\perp} h\|_{\mathcal{S}}^2 \leq \|h\|_{\mathcal{S}}^2. \quad (3.3)$$

For the regulated A1 covariance, mode by mode,

$$C_J(k) = |r_J(k)|^2 \mathcal{A}(k)^{-1}, \quad \mathcal{A}(k) \succeq c_{\text{sym}} \langle k \rangle^4 \mathbf{I}, \quad |r_J(k)| \leq M_R. \quad (3.4)$$

It follows on $\text{Ran } C_J$ that

$$\|L_\pi h\|_{H^2}^2 \leq \frac{M_R^2}{c_{\text{sym}}} \|L_\pi h\|_{q,L_\pi}^2 \leq \frac{M_R^2}{c_{\text{sym}}} \|h\|_{\mathcal{S}}^2. \quad (3.5)$$

For the certified contractive regulator family,

$$M_R = 1, \quad c_{\text{sym}} = 0.10836453058711468 \dots, \quad C_{\text{CM}} = c_{\text{sym}}^{-1} = 9.22811176850986 \dots \quad (3.6)$$

Proof. The Moore–Penrose identity in (3.3) is the standard polar-decomposition identity for L_π . It is used only to evaluate the quotient norm; the control is not replaced by a noncausal minimal representative. On a Fourier mode in the covariance range, (3.4) gives

$$C_J(k)^\dagger \succeq \frac{c_{\text{sym}}}{M_R^2} \langle k \rangle^4 \mathbf{I}.$$

Summation over modes proves (3.5). The statement is independent of the number of visits and of overlap among their ranges. $\square$

Lemma 3.2 (sharp two-endpoint L^6 bridge)

Writing $Z_h = X_J + z$,

$$\|z\|_6^6 \leq 32 \left(\|Z_h\|_6^6 + \|X_J\|_6^6 \right). \quad (3.7)$$

This is the pointwise inequality $|a - b|^6 \leq 2^5(|a|^6 + |b|^6)$, integrated over space. Equality is attained for $b = -a$, so 32 cannot be reduced in this two-term inequality.

Corollary 3.3 (once-only R-118 payment for fixed matrices)

Let $A_0 = M_R^2/c_{\text{sym}}$. Substituting (3.5)–(3.7) into the R-118 operator-valued quadratic estimate gives, for fixed selfadjoint generators with $\sum_r |K_r| \preceq K_0 \mathbf{I}$,

$$\begin{aligned} |\langle Q, \Pi_K z \rangle| &\leq \varepsilon \|h\|_{\mathcal{S}}^2 + \varepsilon_6 (\|Z_h\|_6^6 + \|X_J\|_6^6) \\ &\quad + C\sqrt{32} K_0^{10/3} A_0^{11/6} \varepsilon^{-11/6} \varepsilon_6^{-1/2} \|Q\|_{H^{-11/10}}^{10/3}. \end{aligned} \quad (3.8)$$

Thus horizontal synthesis itself produces neither an inverse singular-value loss nor a visit-count loss. Equation (3.8) is a component theorem, not yet the complete production inequality: the owner-complete Q and generator decomposition must still be proved.

Method boundary 3.4

Quotient geometry without covariance coercivity does not imply (3.5). For $L_N t = t e_N$, with e_N an L^2 -normalized Fourier mode,

$$\|e_N\|_{q,L_N} = 1, \quad \|e_N\|_{H^2}^2 = \langle N \rangle^4.$$

The A8 symbol lower bound is therefore load-bearing.

4. Round three: spatially varying multiplier theorem

For selfadjoint matrix fields $K_r(x)$ assume, for every multi-index $|\gamma| \leq 2$,

$$\sup_x \sum_r |\langle v, \partial^\gamma K_r(x) w \rangle| \leq K_\gamma \|v\| \|w\|, \quad (4.1)$$

and put

$$K_{\leq 2} = K_0 + \sum_{1 \leq |\gamma| \leq 2} K_\gamma.$$

Let $X = \|a\|_{H^2}^2$ and $Y = \|a\|_6^6$.

Theorem 4.1 (two-derivative multiplier extension)

The centered quadratic output $\Pi_K(a)$ satisfies

$$\|\Pi_K(a)\|_2 \leq CK_0 Y^{1/3}, \quad \|\Pi_K(a)\|_{H^2} \leq CK_{\leq 2} X. \quad (4.2)$$

For $\kappa = 1/10$, interpolation at $\theta = (1 + \kappa)/2 = 11/20$ yields

$$\|\Pi_K(a)\|_{H^{11/10}} \leq CK_0^{9/20} K_{\leq 2}^{11/20} X^{11/20} Y^{3/20}. \quad (4.3)$$

Consequently, for $Q \in H^{-11/10}$ and all $\eta, \zeta > 0$,

$$\boxed{|\langle Q, \Pi_K(a) \rangle| \leq \eta X + \zeta Y + CK_0^{3/2} K_{\leq 2}^{11/6} \eta^{-11/6} \zeta^{-1/2} \|Q\|_{H^{-11/10}}^{10/3}} \quad (4.4)$$

For spatially constant matrices, $K_{\leq 2} = K_0$ and the coefficient in the remainder becomes $K_0^{10/3}$, exactly recovering R-118.

Proof. The L^2 estimate follows from the pointwise spectral bound, Holder, and the same sharp-maximal centering used in R-118. Differentiating the quadratic output twice assigns at most two derivatives among a , the second copy of a , and K_r . The summed bilinear bounds (4.1), the algebra property of the finite-cutoff smooth representative, and the R-118 square-function argument give the second estimate in (4.2). Real interpolation gives (4.3). Duality against Q , followed by the same two-parameter Young optimization as R-118, multiplies the interpolation exponents by $10/3$:

$$\frac{9}{20} \frac{10}{3} = \frac{3}{2}, \quad \frac{11}{20} \frac{10}{3} = \frac{11}{6}.$$

This proves (4.4). $\square$

The order-two hypothesis cannot be deleted. With constant a and $K_N(x) = \cos(Nx_1)\mathbf{I}$, the pointwise operator norm stays bounded while the H^2 norm of the quadratic output grows like N^2 . This is an abstract method counterfixture, not a production counterexample.

5. Round four: stationary production low chaoses

R-119 reduces zero/first-chaos cancellation to two aggregate identities. If $Y = b + A\xi + R$, with R orthogonal to zero and first chaos, and $\delta\tau$ denotes the complete trace owner, define

$$D_0 = \mathbb{E}\delta\tau - \mathbb{E}\|R\|^2, \quad (5.1)$$

$$D_1 = \mathbb{E}D\delta\tau - 2\text{Contr}(A, \mathbb{E}D^2R) - 2\mathbb{E}[(DR)^*R]. \quad (5.2)$$

The low chaoses vanish exactly when $D_0 = D_1 = 0$.

Proposition 5.1 (exact one-pair production checksum)

For the actual first linear row, let

$$z = \frac{\sigma}{2}(\xi - i\eta), \quad \xi, \eta \sim N(0, 1), \quad \lambda_1 = 4c_0 = \frac{6}{125P}.$$

The square-minus-trace diagnostic is

$$\mathcal{P} = \frac{\lambda_1 \sigma^4}{16} \left[(\xi^2 + \eta^2)^2 - 4(\xi^2 + \eta^2) \right]. \quad (5.3)$$

Its exact Hermite expansion is

$$\mathcal{P}_2 = \frac{\lambda_1 \sigma^4}{4} \{H_2(\xi) + H_2(\eta)\}, \quad (5.4)$$

$$\mathcal{P}_4 = \frac{\lambda_1 \sigma^4}{16} \{H_4(\xi) + H_4(\eta) + 2H_2(\xi)H_2(\eta)\}. \quad (5.5)$$

Therefore $\Pi_0 \mathcal{P} = \Pi_1 \mathcal{P} = 0$. Equivalently,

$$\mathbb{E}\tau = \mathbb{E}\|Y\|^2 = \lambda_1 \sigma^4, \quad \mathbb{E}D\tau = 2\mathbb{E}[(DY)^*Y] = 0. \quad (5.6)$$

Theorem 5.2 (stationary six-row A1 raw-current pass)

Let U be the complete fixed-cutoff stationary A1 Gaussian field with a common entrywise-real-even scalar regulator. For each spatial direction put

$$Y_i = C(U)\partial_i U, \quad \tau_i = \text{Tr}(C(U)^T C(U)\Gamma_i), \quad (5.7)$$

where C contains all six rows in (2.3) and Γ_i is the derivative covariance. Then the complete stationary packet satisfies both R-119 low-chaos identities:

$$\mathbb{E}\tau = \mathbb{E}\|Y\|^2, \quad \mathbb{E}D\tau = 2\mathbb{E}[(DY)^*Y] = 0. \quad (5.8)$$

Proof. Real-even stationarity gives

$$\mathbb{E}[U \otimes \partial_i U] = 0.$$

Because the pair is jointly Gaussian, U and $\partial_i U$ are independent. Conditioning on U therefore proves

$$\mathbb{E}[\|Y_i\|^2 \mid U] = \text{Tr}(C(U)^T C(U)\Gamma_i) = \tau_i. \quad (5.9)$$

Every q_A is even and every coefficient row is odd, so $C(-U) = -C(U)$. Under simultaneous sign reversal of the complete Gaussian root, both Y_i and τ_i are even. Hence their zero/first-chaos decompositions have $b = 0$, $A = 0$, $\mathbb{E}D\tau = 0$, and $\mathbb{E}[(DY)^*Y] = 0$; the last two integrands are odd. Summing over directions and rows proves (5.8). $\square$

This result includes all six raw-current rows, including the rational rows. It is stronger than the one-pair checksum, but it is not an owner-complete trace/heat/low/R-063 forest theorem and is deliberately not stated for an adapted triangular shift.

Adapted boundary 5.3

In the legal two-block chart, the second control is $h_2(\xi_1)$. The root-one chain rule then contains owner families of the form

$$DB[Dh_2], \quad D^2B[DX, Dh_2], \quad D^2B[Dh_2, Dh_2], \quad DB[D^2h_2]. \quad (5.10)$$

Neither stationary independence nor global parity determines their sum. R-104 proves that some strict-triangular factorization exists, but it does not specify canonical coefficient-level S_1, S_2, h_1, h_2 ; R-063 controls deterministic H^2 translations, not this adapted random shift. Therefore the production values of (5.1)–(5.2) remain open. This is now the minimal low-chaos target: no higher-chaos expansion is needed until D_0, D_1 are computed.

6. Round five: linear terminal Hessian

For the linear rows define

$$\mu_A = 2u^T S_A z, \quad n_A = z^T S_A z.$$

The covariance-normal linear functional is

$$V_L^{\text{ren}}(u) = \frac{c_0}{2} \sum_{A,i} \int \left\{ (\partial_i m_A)^2 - 4(S_A u)^T \Gamma_i(S_A u) \right\}. \quad (6.1)$$

Theorem 6.1 (exact linear-row Hessian)

For every smooth cylindrical direction z ,

$$\boxed{D^2 V_L^{\text{ren}}(u)[z, z] = c_0 \sum_A \|\nabla \mu_A\|_2^2 - 2c_0 \sum_A \langle \Delta m_A, n_A \rangle - 4c_0 \sum_{A,i} \int (S_A z)^T \Gamma_i(S_A z)}. \quad (6.2)$$

The first line is a positive square plus a fixed-selfadjoint centered quadratic coefficient; the second line is exactly the once-owned covariance trace. Since $J_A = \text{dm}_A$ is exact, its exterior curvature vanishes. The linear rows cannot supply a rational Cartan companion.

The Taylor remainder of the quartic current uses

$$\int_0^1 (1-t) 3t^2 dt = \frac{1}{4}, \quad (6.3)$$

and its mixed quadratic products split with powers $1/2 + 1/2 = 1$. These checks prevent a hidden factor-two or extra sextic payment.

7. General current Hessian and rational Cartan survivor

Let $\theta = f(u) \cdot du$ be a one-form and set

$$j_i = \theta(\partial_i u), \quad \sigma = \theta(z), \quad \tau = D\thetaz, \quad \Omega = d\theta.$$

The first variation has the invariant form

$$\dot{j}_i = \partial_i \sigma + \Omega(z, \partial_i u). \quad (7.1)$$

Theorem 7.1 (current-Hessian identity)

On the torus,

$$D^2 \frac{1}{2} \sum_i \|j_i\|_2^2 = \sum_i \int \left\{ |\partial_i \sigma + \Omega(z, \partial_i u)|^2 - (\partial_i j_i) \tau + j_i [(D_z \Omega)(z, \partial_i u) + \Omega(z, \partial_i z)] \right\}. \quad (7.2)$$

The covariance-normal trace contributes exactly

$$- \sum_i \int \left\{ (Df[z])^T \Gamma_i Df[z] + (D^2 f[z, z])^T \Gamma_i f \right\}. \quad (7.3)$$

Equations (7.2)–(7.3) retain all sign and factor information before any analytic estimate.

For the rational row

$$\theta_A = dm_A - \alpha q_A d\rho,$$

the curvature is

$$\boxed{\Omega_A = d\theta_A = -\frac{\alpha}{d} dm_A \wedge d\rho.} \quad (7.4)$$

Thus the terminal Hessian contains the first-order Cartan block

$$\mathcal{C}_A(z) = \sum_i \int j_{A,i} [(D_z \Omega_A)(z, \partial_i u) + \Omega_A(z, \partial_i z)]. \quad (7.5)$$

It vanishes on special radial loci but is generically nonzero. No zeroth-order fixed-matrix theorem can silently replace (7.5).

8. Rational raw-covariance Hessian and fixed basis

Define

$$T_A = S_A - \alpha q_A \mathbf{I}, \quad r_A = (S_A - q_A \mathbf{I})u, \quad g_A = \nabla q_A = \frac{2r_A}{d}, \quad (8.1)$$

$$H_A = D^2 q_A = \frac{2(S_A - q_A \mathbf{I})}{d} - \frac{4}{d^2} (u \otimes r_A + r_A \otimes u), \quad (8.2)$$

and

$$F_A = T_A - \alpha u \otimes g_A. \quad (8.3)$$

For a symmetric rough matrix Q , set

$$w_A = QT_A u, \quad \beta_A = u^T Q T_A u.$$

Theorem 8.1 (exact rational zeroth-order Hessian)

The rational raw- Q contribution satisfies

$$\frac{1}{2} D^2 B_R(u)[z, z] : Q = c_1 \sum_A z^T \mathcal{K}_A(Q) z, \quad (8.4)$$

where

$$\boxed{\mathcal{K}_A(Q) = 4F_A^T Q F_A - 4\alpha \beta_A H_A - 4\alpha (g_A \otimes w_A + w_A \otimes g_A).} \quad (8.5)$$

Each $\mathcal{K}_A(Q)$ is selfadjoint. Formula (8.5) is obtained by differentiating $4c_1 (T_A u)^T Q (T_A u)$ twice; no derived coefficient is pasted.

Every symmetric six-real matrix has the fixed expansion in

$$K_{pp} = e_p e_p^T, \quad K_{pq} = \frac{1}{2} (e_p e_q^T + e_q e_p^T), \quad p < q. \quad (8.6)$$

There are $6 \cdot 7/2 = 21$ such matrices, and

$$\boxed{\sum_{p \leq q} |K_{pq}| = \frac{7}{2} \mathbf{I}.} \quad (8.7)$$

Hence the matrix direction of the zeroth-order rational Hessian is flattened with a cutoff- and visit-independent absolute spectral constant.

This does not finish the production multiplier estimate. The scalar coefficients in (8.5) depend on u, q_A, g_A, H_A, Q and are random under an adapted translation. R-063's deterministic-translation estimate does not yet certify the derivative bounds (4.1), and the separate first-order block (7.5) is untouched by (8.6)–(8.7).

9. Cartan checksum and exact surviving frontier

The R-102 active/kernel fixture has isolated rational curl

$$\partial_y \omega_x - \partial_x \omega_y = -\frac{40}{729}. \quad (9.1)$$

Complete terminal-action exactness therefore requires the complementary owner sum to contribute

$$\text{curl } \omega_{\text{companion}} = +\frac{40}{729}. \quad (9.2)$$

The three linear rows have zero curvature and cannot provide (9.2). The companion must be reconstructed from the remaining rational terminal square, covariance trace, heat, low-output terms, and the R-063 Wick forest, each owned exactly once. R-120 reproduces (9.1) and derives the requirement (9.2); it does not report the companion as observed.

The new frontier is therefore smaller than the R-119 checklist:

1. reconstruct the legal two-block/two-visit rational owner sum at coefficient level;
2. compute only D_0, D_1 from (5.1)–(5.2), including every family in (5.10);
3. on the R-102 slice, verify the independent checksum (9.2);
4. either cancel the Cartan block (7.5) inside that complete owner or prove a direct one-use bound for it;
5. certify (4.1) for the residual scalar multipliers, then invoke (3.8)/(4.4) once;
6. only after that run the global A13 one-use, $\text{OVERLAP}_{\text{src}}$, Nelson, and removal gates.

10. Evidence map: successful and failed routes

1. **Covariance quotient route – advanced and closed at this layer.** R-092/R-104 quotient geometry plus A8 symbol coercivity proves (3.5); (3.7) supplies the missing terminal sextic bridge. Revisit count and range overlap introduce no new constant.
2. **Spatially constant multiplier route – retained.** R-118 and Corollary 3.3 give the exact once-only estimate for fixed matrices.
3. **Spatially varying multiplier route – conditionally advanced.** Theorem 4.1 proves the analytic extension under (4.1). Production coefficients have not yet been shown to satisfy those derivative bounds.
4. **One-pair low-chaos route – passed but diagnostic.** Proposition 5.1 is an actual A1 linear row and catches normalization errors; it is not the whole adapted cluster.
5. **Complete stationary six-row raw-current route – passed.** Theorem 5.2 includes the rational rows. It does not reconstruct the trace/heat/low/R-063 forest. Its independence/parity mechanism is destroyed by general adapted future feedback, so it cannot close D_0, D_1 .
6. **Fixed-matrix Hessian route – partly successful.** The linear rows close exactly, and the zeroth-order rational block has the 21-matrix flattening. The Cartan block is first order and survives.
7. **Isolated-current primitive route – failed previously and not revived.** The nonzero curl (9.1) prevents a primitive for the isolated rational current. Only a complete-owner companion can repair it.

8. **Stationary-to-adapted parity transfer – rejected.** The families (5.10) are genuine and have not been assigned coefficients. A stationary pass is evidence for signs and conventions, not a proof of the adapted packet.

11. Executable verification

The primary SymPy certificate reconstructs the A1 diagonalization from its upstream constants, the fixed Pauli identities, a nonorthogonal covariance quotient fixture, the A8 coercivity constant, the sharp L^6 bridge, all multiplier exponents, the exact one-pair Hermite split, stationary parity/conditional trace, the linear Hessian and trace factor, rational curvature, the R-102 curl, three exact raw- Q Hessian fixtures, the fixed 21-matrix absolute sum, and the adapted chain-rule families. It passes 68/68 assertions.

A non-importing standard-library certificate uses exact rational arithmetic and a separate matrix implementation. It independently reconstructs the production constants and realified generators, covariance quotient, Gaussian moments, all six coefficient-row parities, multiplier exponents, linear completion, rational Hessian, the R-102 curl, the explicitly assembled fixed basis, and the adapted chain rule. Its exact assertion count is pinned by the manifest; the present package passes 59/59. The integrated verifier reruns both children into temporary outputs, compares deterministic payloads, validates every authority/source/note/PDF hash and contract field, and fails closed on stale evidence.

12. Devil’s-advocate review

1. **Objection: quotient unitarity alone proves the H^2 bound. UPHELD as false.** Method boundary 3.4 has unit quotient norm and unbounded H^2 norm. The A8 coercivity constant is essential.
2. **Objection: the Moore–Penrose representative violates causality. DISMISSED.** It is used only to identify the quotient norm. Equation (3.3) bounds the causal supplied control h by orthogonal projection and never replaces h in the dynamics.
3. **Objection: a pointwise spectral bound controls a varying multiplier in H^2 . UPHELD as false.** The oscillatory $K_N(x)$ fixture forces derivative growth. Bounds through order two in (4.1) are load-bearing.
4. **Objection: the stationary A1 pass proves adapted low-chaos cancellation. UPHELD as false.** Adapted $h_2(\xi_1)$ generates the four unresolved chain-rule families (5.10). The exact remaining data are D_0, D_1 , not a generic parity statement.
5. **Objection: the one-pair computation hides the rational rows. VALID-with-mitigation.** Proposition 5.1 is explicitly diagnostic. Theorem 5.2 separately covers the stationary six-row raw-current packet, including rational rows, by conditional Gaussianity and parity; it does not claim the adapted trace/heat/low/R-063 forest.
6. **Objection: the 21-matrix basis removes every rational Hessian difficulty. UPHELD as false.** It flattens only the zeroth-order matrix direction. The adapted scalar multipliers and the first-order Cartan block remain.
7. **Objection: terminal-action exactness guarantees the +40/729 companion without computation. VALID-with-mitigation.** It makes (9.2) a necessary checksum, but cannot validate owner signs or multiplicities. The companion is deliberately recorded as required and not observed.
8. **Objection: derived constants or factors may be hardcoded. VALID-with-mitigation.** Both children read the A1/A8 upstream artefacts, derive $c_0, c_1, \alpha, c_{\text{sym}}$, and reconstruct all exponent

relations. The independent child uses exact Gaussian moment and matrix routines. Decimal values appear only as regression oracles after derivation.

9. **Objection: the result closes either A13 gate or Sector A. DISMISSED.** The adapted owner reconstruction, D_0, D_1 , Cartan companion/bound, complete one-use inequality, $\text{OVERLAP}_{\text{src}}$, Nelson, and removals remain open. Tier stays T4.

13. Result footer

Result ID: A13-CLASSII-COVARIANCE-HORIZONTAL-SYNTHESIS-STATIONARY-
LOW-CHAOS-CARTAN-HESSIAN-BOUNDARY (R-120).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: Thm. 2.1; Thm. 3.1, Lem. 3.2, Cor. 3.3;
Thm. 4.1; Prop. 5.1, Thm. 5.2, boundary 5.3; Thms. 6.1, 7.1,
8.1; checksum (9.1)-(9.2); frontier and evidence map.
Scope: fixed finite cutoff and positive rational floor; A1 six-real frame;
stationary complete low-chaos result is not an adapted result.
Dependencies: A1, A8, R-063,068,082,089,092,102,104,107,116,118,119.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_covariance_horizontal_synthesis_stationary_
low_chaos_cartan_hessian_boundary_verify.py
Expected output: R-120 PASS; child and integrated counts pinned by manifest;
all hashes and no-overclaim boundaries valid.
Falsification gate: any failed coefficient, covariance, multiplier, chaos,
Hessian, Cartan, basis, adapted-boundary, hash, or PDF assertion.
Tier before / after: T4 / T4.
No-overclaim statement: adapted D0/D1, Cartan companion, complete one-use,
 $\text{OVERLAP}_{\text{src}}$, Nelson, removals, measure, Sector-A/T5-T7 remain open.
Next required action: reconstruct the complete two-block rational owner sum,
compute D0,D1, and test the independent +40/729 companion checksum.